



# Caso 3: Cloud Native

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Agustín Montaña

Legajo: 60160

Universidad de Mendoza

Ing. Informática

Teleinformática

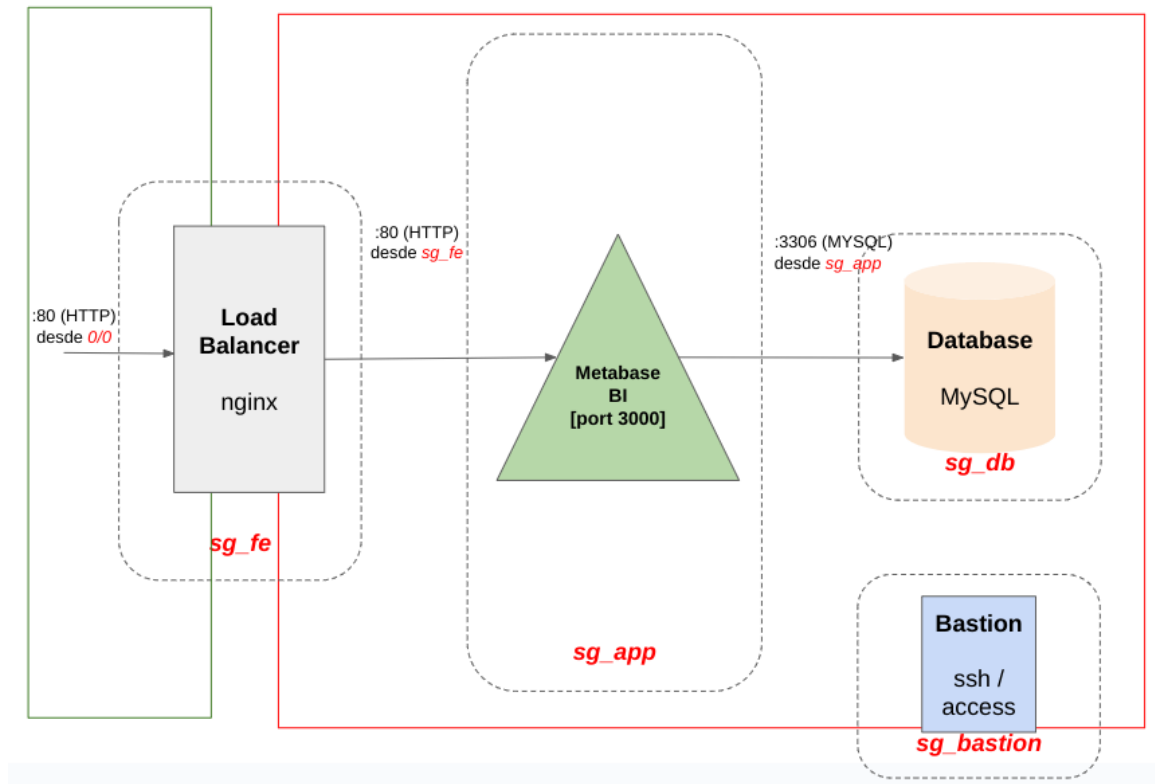
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## Enunciado

- Desplegar la plataforma de BI Metabase en la arquitectura planteada a continuación (en Kubernetes).
- Utilizar el dataset provisto de Google Mobility y generar una visualización como la planteada.



## Recursos

- [Metabase](#)
- [Google Mobility Reports](#)

## Resolución

1. Preparacion de recursos:
  - a. Cree una instancia small con flavor um-kube-tools

# amontana\_kube

Overview

Interfaces

Log

Console

Action Log

Name

ID

Description

Project ID

Status

Locked

Availability Zone

Created

Age

amontana\_kube

e06c4128-11b3-4456-8cdd-2c194814d9b6

-

c6df3923078b40259b7575a0a6e17913

Active

False

nodos-amd-2022

Aug. 2, 2023, 3:18 p.m.

6 days, 7 hours

Specs

Flavor Name

Flavor ID

RAM

VCPUs

Disk

m1.small

2d357d3d-32c1-4af8-81dd-a71a7d7cf303

2GB

1 VCPU

20GB

IP Addresses

net\_umstack

10.201.3.38, 2002:c833:29b0:2:f816:3eff:fe22:1264

Security Groups

default

ALLOW IPv4 from default

ALLOW IPv6 to ::/0

ALLOW IPv6 from default

ALLOW IPv4 icmp from 0.0.0.0/0

ALLOW IPv4 22/tcp from 0.0.0.0/0

ALLOW IPv4 to 0.0.0.0/0

ALLOW IPv4 5900-5910/tcp from 0.0.0.0/0

Metadata

Key Name

Image Name

Image ID

amontana\_acer

um-kube-tools

2e1a3e91-834a-4435-8d20-887b9d53de31

b. Ingreso a la instancia con `ssh -L 8080:localhost:8080 -L 8081:localhost:8081 -A ubuntu@10.201.3.38`

c. Segui los pasos de `um-help` para iniciar sesión en Rancher y crear el namespace `amontana-dev`

## 2. Copiado de el archivo google-mobility.sql

```
scp google-mobility.sql ubuntu@10.201.3.38:/Github/Teleinformatica/Caso_3/
```

## 3. Archivos YAML de MySQL:

a. Secret

```

apiVersion: v1
kind: Secret
metadata:
  creationTimestamp: null
  name: meta-db-secrets
type: Opaque
data:
  MYSQL_ROOT_PASSWORD: bXlzcWwxMjM0
  MYSQL_DATABASE: bWV0YWJhc2UtZGI=

```

## b. ConfigMap

```

apiVersion: v1
kind: ConfigMap
metadata:
  name: mysql-config
data:
  my.cnf: |
    [mysqld]
    default-authentication-plugin=mysql_native_password

```

## c. StatefulSet

```

apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: metabase-db
  labels:
    run: metabase-db
spec:
  replicas: 1
  selector:
    matchLabels:
      run: metabase-db
  serviceName: "mysql-service"
  template:
    metadata:
      generateName: metabase-db
      labels:
        run: metabase-db
    spec:
      initContainers:
        - name: "remove-lost-found"
          image: "busybox:1.25.0"
          imagePullPolicy: "IfNotPresent"
          command: ["rm", "-fr", "/var/lib/mysql/lost+found"]
          volumeMounts:
            - name: mysql-data
              mountPath: /var/lib/mysql
      containers:
        - name: mysql
          image: mysql:5.7
          imagePullPolicy: IfNotPresent
          env:
            - name: MYSQL_DATABASE
              valueFrom:
                secretKeyRef:
                  name: meta-db-secrets
                  key: MYSQL_DATABASE
            - name: MYSQL_ROOT_PASSWORD
              valueFrom:
                secretKeyRef:
                  name: meta-db-secrets

```

```

        key: MYSQL_ROOT_PASSWORD
    ports:
      - containerPort: 3306
    volumeMounts:
      - name: mysql-data
        mountPath: /var/lib/mysql
      - name: mysql-config
        mountPath: /etc/mysql/conf.d/my.cnf
        subPath: my.cnf
      - name: mysql-volume
        mountPath: /docker-entrypoint-initdb.d
    volumes:
      - name: mysql-config
        configMap:
          name: mysql-config
      - name: secret-volume
        secret:
          secretName: meta-db-secrets
  volumeClaimTemplates:
    - metadata:
        name: mysql-data
      spec:
        accessModes: [ "ReadWriteOnce" ]
        storageClassName: longhorn
        resources:
          requests:
            storage: 2Gi
    - metadata:
        name: mysql-volume
      spec:
        accessModes: [ "ReadWriteOnce" ]
        storageClassName: longhorn
        resources:
          requests:
            storage: 2Gi

```

#### d. Service

```

apiVersion: v1
kind: Service
metadata:
  name: mysql-service
  labels:
    run: metabase-db
spec:
  ports:
    - port: 3306
      name: metabase-db
  selector:
    run: metabase-db

```

### 4. Archivos YAML de Metabase:

#### a. Secret

- i. TYPE → mysql
- ii. DBNAME → metabase-db
- iii. PORT → 3306
- iv. USER → root
- v. HOST → mysql-service

## vi. PASS → mysql1234

```
apiVersion: v1
kind: Secret
metadata:
  name: metabase-secrets
type: Opaque
data:
  MB_DB_TYPE: "bXlzcWw="
  MB_DB_DBNAME: "bWV0YWJhc2UtZGI="
  MB_DB_PORT: "MzMwNg=="
  MB_DB_USER: "cm9vdA=="
  MB_DB_HOST: "bXlzcWwtc2VydmljZQ=="
  MB_DB_PASS: bXlzcWwzMjM0
```

## b. Service

```
apiVersion: v1
kind: Service
metadata:
  name: metabase
  labels:
    run: metabase
spec:
  selector:
    run: metabase
  ports:
    - name: metabase
      port: 3000
      targetPort: 3000
  type: ClusterIP
```

## c. Deployment

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: metabase
  labels:
    run: metabase
spec:
  replicas: 1
  selector:
    matchLabels:
      run: metabase
  template:
    metadata:
      generateName: metabase
      labels:
        run: metabase
    spec:
      containers:
        - name: metabase
          image: metabase/metabase:latest
          imagePullPolicy: Always
          envFrom:
            - secretRef:
                name: metabase-secrets
          ports:
            - name: metabase
              containerPort: 3000
          volumeMounts:
            - name: metabase-secrets
```

```

        mountPath: /metabase-secrets
      volumes:
      - name: metabase-secrets
        secret:
          secretName: metabase-secrets

```

#### d. Ingress

```

apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: metabase-ingress
  annotations:
    nginx.ingress.kubernetes.io/rewrite-target: /
    kubernetes.io/ingress.class: nginx
spec:
  rules:
  - host: amontana-metabase.my.kube.um.edu.ar
    http:
      paths:
      - path: /
        pathType: Prefix
        backend:
          service:
            name: metabase
            port:
              number: 3000

```

#### 5. Hago apply de todo:

```
kubectl apply -f .
```

| NAME                          | READY | STATUS  | RESTARTS | AGE  |
|-------------------------------|-------|---------|----------|------|
| pod/kb-prod-0                 | 1/1   | Running | 0        | 8d   |
| pod/metabase-7459d764cb-rksnm | 1/1   | Running | 0        | 172m |
| pod/metabase-db-0             | 1/1   | Running | 0        | 172m |

| NAME                  | TYPE      | CLUSTER-IP    | EXTERNAL-IP | PORT(S)  | AGE  |
|-----------------------|-----------|---------------|-------------|----------|------|
| service/kb-prod       | ClusterIP | 10.43.188.226 | <none>      | 80/TCP   | 41d  |
| service/metabase      | ClusterIP | 10.43.77.223  | <none>      | 3000/TCP | 177m |
| service/mysql-service | ClusterIP | 10.43.133.241 | <none>      | 3306/TCP | 172m |

| NAME                     | READY | UP-TO-DATE | AVAILABLE | AGE  |
|--------------------------|-------|------------|-----------|------|
| deployment.apps/metabase | 1/1   | 1          | 1         | 172m |

| NAME                                | DESIRED | CURRENT | READY | AGE  |
|-------------------------------------|---------|---------|-------|------|
| replicaset.apps/metabase-7459d764cb | 1       | 1       | 1     | 172m |

| NAME                         | READY | AGE  |
|------------------------------|-------|------|
| statefulset.apps/kb-prod     | 1/1   | 41d  |
| statefulset.apps/metabase-db | 1/1   | 172m |

- Modifique el archivo `google-mobility.sql` para que se cree un database “mobility” al iniciar MySQL y se rellene con el contenido de `docker-entrypoint-initdb.d`

```

1  -- MySQL dump 10.16  Distrib 10.1.48-MariaDB, for debian-linux-gnu (x86_64)
2  --
3  -- Host: 10.201.0.43    Database: google
4  -- -----
5  -- Server version 5.7.41-0ubuntu0.18.04.1
6
7  /*!40101 SET @OLD_CHARACTER_SET_CLIENT=@@CHARACTER_SET_CLIENT */;
8  /*!40101 SET @OLD_CHARACTER_SET_RESULTS=@@CHARACTER_SET_RESULTS */;
9  /*!40101 SET @OLD_COLLATION_CONNECTION=@@COLLATION_CONNECTION */;
10 /*!40101 SET NAMES utf8mb4 */;
11 /*!40103 SET @OLD_TIME_ZONE=@@TIME_ZONE */;
12 /*!40103 SET TIME_ZONE='+00:00' */;
13 /*!40014 SET @OLD_UNIQUE_CHECKS=@@UNIQUE_CHECKS, UNIQUE_CHECKS=0 */;
14 /*!40014 SET @OLD_FOREIGN_KEY_CHECKS=@@FOREIGN_KEY_CHECKS, FOREIGN_KEY_CHECKS=0 */;
15 /*!40101 SET @OLD_SQL_MODE=@@SQL_MODE, SQL_MODE='NO_AUTO_VALUE_ON_ZERO' */;
16 /*!40111 SET @OLD_SQL_NOTES=@@SQL_NOTES, SQL_NOTES=0 */;
17
18 --
19 -- Table structure for table `mobility`
20 --
21
22 CREATE DATABASE IF NOT EXISTS mobility;
23 USE mobility;
24
25 DROP TABLE IF EXISTS `mobility`;
26 /*!40101 SET @saved_cs_client      = @@character_set_client */;
27 /*!40101 SET character_set_client = utf8 */;

```

7. Copiar archivo google-mobility.sql a la carpeta `docker-entrypoint-initdb.d` la cual almacena los archivos que se ejecutan al iniciar un contenedor de MySQL u otros en un pod de Kubernetes. En los YAML especifique que la carpeta debe persistir en un volumen, por lo que el archivo no debería perderse inclusive eliminado el pod.

```
kubectl cp ./google-mobility.sql metabase-db-0:/docker-entrypoint-initdb.d/google-mobility.sql -c mysql
```

8. Reinicio del pod y recursos para cargar la DB.

```
kubectl delete pod <nombre-del-pod>
kubectl delete ...
```

**NOTA:** hice un script en python que iniciara, borrara o reiniciará los recursos por mi, ya que tuve que hacerlo varias veces, esta en el zip y en la carpeta de Caso\_3 en la instancia. El comando que sue fue:

```
python3 restart.py -restart
```

9. Verificar que los datos se cargaran correctamente en la DB:

```

# Acceso al pod
kubectl exec -it metabase-db-0 -- bash

# Acceso a mysql
mysql -u root -p

# Verificar datos

```

```
SHOW DATABASES;  
USE mobility;  
SHOW TABLES;  
SELECT * FROM mobility LIMIT 10;
```

Con eso verifique que los datos estuvieran bien cargados.

10. Verificar que Metabase se ejecute correctamente y se conecte a la DB.

```
kubectl logs <nombre-del-pod>
```

11. Acceso a Metabase y configuración de la pagina:

**URL:** [amontana-metabase.my.kube.um.edu.ar](http://amontana-metabase.my.kube.um.edu.ar)

MySQL



Nombre para mostrar

Metabase Kubernetes



Host

mysql-service



Port

3306

Database name

mobility

Username

root

Password

●●●●●●●●

Use a secure connection (SSL)



Use an SSH-tunnel



Si no es posible una conexión directa a tu base de datos, es posible que quieras utilizar un túnel SSH. [Aprende más.](#)

Mostrar opciones avanzadas ▼

Metabase Kubernetes / Mobility



Filtro

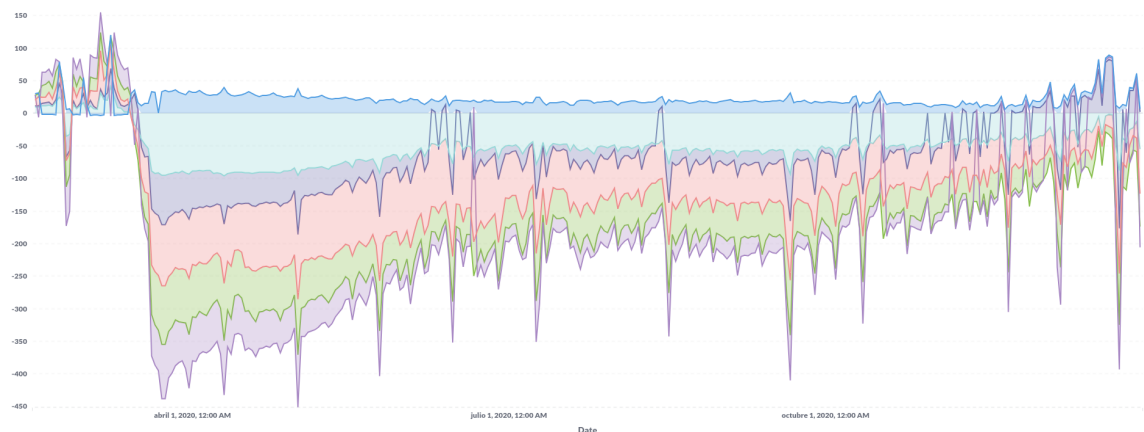
Resumir



Guardar

Date entre enero 1, 2020 diciembre 31, 2020 X Sub Region 1 es Mendoza Province X Sub Region 2 es Capital Department X

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Visualización



